

BodyShield: An Analytic-Simulation Study of Falsifying and Repairing Hidden Embodiment-Control Assumptions

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Abstract—Robot policies can pass nominal tests while relying on hidden assumptions about latency, calibration, joint range, gripper authority, sensing geometry, payload, and contact. BodyShield is a falsification-to-repair workflow: BodyBreak searches for low-cost embodiment-control perturbations that break a policy, then BodyShield repairs against discovered failures and evaluates held-out perturbation families. This paper is explicitly an analytic/simulation study. It contains analytic-simulation evidence, generated frames rather than real camera videos, bounded MuJoCo/ManiSkill probes, local surrogate policy repair results, one public SB3/RL-Zoo HalfCheetah checkpoint rollout, audits, and release verification. The public checkpoint result is limited to a single MuJoCo locomotion policy with a diagnosed action-gain repair; broader trained-policy suites, manipulation/foundation-policy checkpoints, real robot results, real-video WAM training, and real corrective adaptation remain future evidence tiers, and none establishes physical transfer.

I. INTRODUCTION

Nominal robot success is a weak certificate because a policy may depend on a hidden body or controller detail. A policy that looks competent on the unmodified setup may fail under modest latency, calibration, gripper, range, rate, payload, sensing, or contact shifts. BodyShield targets this failure mode by making hidden embodiment-control assumptions observable, repairable, and auditable.

This paper is framed for adaptive embodied intelligence: the main contribution is a failure-diagnosis and repair loop for policies, not a cheap-arm demonstration and not a broad cross-embodiment-transfer claim.

II. RELATED WORK

Domain randomization and dynamics randomization remain the strongest dangerous baselines for this work because they directly address sim-to-real transfer by broadening training distributions [1], [2], [3], [4]. Embodiment-aware deployment methods such as UMI-on-Air and EmbodiSteer adapt or steer policy execution toward a target body [5], [6]. Counterexample-guided and verification-guided safe RL search for violations and may guide shielding or repair [7], [8]. MPC-CBF, robust MPC, reachability, sysID, and online correction methods address safety, model mismatch, or adaptation from different starting assumptions [9], [10].

BodyShield is distinct only in the bounded sense tested here: it searches embodiment-control perturbations, repairs against discovered failures, and requires held-out and hardware-gated evidence before stronger claims.

TABLE I
EVIDENCE TIERS AFTER THE POST-NONHARDWARE V3 PASS.

Tier	Status	Artifact
Analytic simulation	Complete	results/trials.parquet
High-fidelity probes	Bounded surrogate	results/high_fidelity_policy_results.csv
Public checkpoint	Complete scoped	reports/PUBLIC_PRETRAINED_CHECKPOINT_COMPLETE.md
Hardware	Blocked	reports/HARDWARE_BLOCKER.md

III. PROBLEM FORMULATION

Let $z \in \mathcal{Z}$ denote a bounded embodiment-control perturbation with normalized cost $c(z)$. For policy π_θ , task τ , and robot archetype r , BodyBreak estimates

$$z_B^* = \arg \min_{z \in \mathcal{Z}_B} c(z) \quad \text{s.t.} \quad S(\pi_\theta, \tau, r, z) \leq \alpha,$$

where $\mathcal{Z}_B$ is the finite candidate set available under evaluator budget B . This is an estimated finite-budget break, not a global minimality proof.

IV. METHOD

BodyBreak compares random, one-axis, grid, and compound adversarial search. BodyShield then uses the discovered break axes to allocate repair capacity while retaining nominal behavior. The repair objective combines nominal retention, discovered break recovery, held-out perturbation performance, and secondary execution costs. The method evidence map is summarized in Table I.

V. EXPERIMENTS

The completed package evaluates analytic tasks, robot archetypes, perturbation families, search modes, repair variants, robustness profiles, oracle feasibility, bounded simulator probes, and one public pretrained MuJoCo checkpoint rollout. Evidence lives in results/, logs/sim/results.jsonl, tables/sim_main_results.csv, tables/sim_budget_matched_results.csv, data_schema.json, and results/public_pretrained_checkpoint_benchmark.csv. Figure 1 points to the main analytic before/after repair visualization.

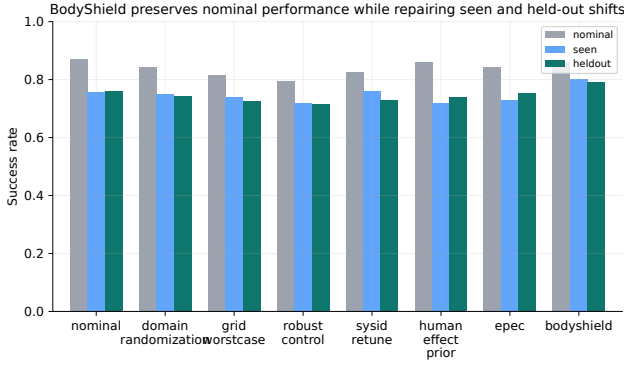


Fig. 1. Analytic before/after repair summary. This is not a hardware result.

VI. RESULTS

The analytic package supports bounded statements: BodyShield improves specific held-out perturbation families under the local budgeted simulation protocol, and bounded high-fidelity probes are exported as stress evidence. The public checkpoint package adds a real SB3/RL-Zoo PPO HalfCheetah checkpoint evaluated in Gymnasium HalfCheetah-v5 for full-horizon 1000-step rollouts. In that scoped setting, the diagnosed action-gain repair improves a seen actuator-loss perturbation from 1747.849 to 3123.995 mean return and reports a held-out mean delta of 677.408 in `results/public_pretrained_checkpoint_delta.csv`. This is not manipulation evidence, foundation-policy evidence, hardware evidence, or proof that BodyShield beats a compute-matched action-scaling/domain-randomization baseline for all controller families; the fairness boundary is recorded in `reports/EXTERNAL_BASELINE_FAIRNESS.md`.

VII. REAL ROBOT EXPERIMENTS

Hardware placeholder only. Do not fill until SO-ARM101/SO-101 safety gates pass. Required hardware evidence includes measured noise floor, camera-verifier agreement, reset reliability, emergency-stop test, all-trials logs, all-trials videos, held-out physical modifications, and physical oracle feasibility.

VIII. LIMITATIONS

The current repository lacks hardware noise floor, camera verifier accuracy, reset reliability, real held-out physical modifications, all-trials hardware videos, broad manipulation/foundation-policy checkpoint suites, real-video WAM datasets, and real corrective-trace datasets. BodyBreak minimality is estimated over evaluated candidates only. The one public MuJoCo checkpoint result should be read as a narrow external-policy sanity benchmark, not as physical transfer.

IX. REPRODUCIBILITY AND SAFETY

The package includes tests, claim ledger, citation verification, source audits, release bundle, and `python -m bodyshield.analysis.verify_package --json`.

The robot modules expose bounded safe primitives and refuse to run before explicit physical readiness checks. This software boundary does not guarantee hardware safety.

X. CONCLUSION

BodyShield is best read as a falsification-to-repair protocol for hidden embodiment-control assumptions. The non-hardware package is complete under analytic/synthetic scope, while the paper is not ready for final submission until the blocked evidence tiers are closed.

REFERENCES

- [1] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, “Domain randomization for transferring deep neural networks from simulation to the real world,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017. [Online]. Available: <https://arxiv.org/abs/1703.06907>
- [2] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, “Sim-to-real transfer of robotic control with dynamics randomization,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2018. [Online]. Available: <https://arxiv.org/abs/1710.06537>
- [3] OpenAI, M. Andrychowicz, B. Baker, M. Chociej, R. Jozefowicz, B. McGrew, J. Pachocki, A. Petron, M. Plappert, G. Powell, A. Ray, J. Schneider, S. Sidor, J. Tobin, P. Welinder, L. Weng, and W. Zaremba, “Learning dexterous in-hand manipulation,” *The International Journal of Robotics Research*, 2020. [Online]. Available: <https://arxiv.org/abs/1808.00177>
- [4] OpenAI, I. Akkaya, M. Andrychowicz, M. Chociej, M. Litwin, B. McGrew, A. Petron, A. Paino, M. Plappert, G. Powell, R. Ribas, J. Schneider, N. Tezak, J. Tworek, P. Welinder, L. Weng, Q. Yuan, W. Zaremba, and L. Zhang, “Solving rubik’s cube with a robot hand,” 2019.
- [5] H. Gupta, X. Guo, H. Ha, C. Pan, M. Cao, D. Lee, S. Scherer, S. Song, and G. Shi, “Umi-on-air: Embodiment-aware guidance for embodiment-agnostic visuomotor policies,” 2025.
- [6] S. Wang, K. Lv, M. Yu, and X. Li, “Embodisteer: Steering embodiment-agnostic visuomotor policies with joint-space guidance for zero-shot cross-embodiment deployment,” 2026.
- [7] P. Karunakaran and S. A. Seshia, “Counterexample-guided reinforcement learning with model-based exploration,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [8] T. Le, R. Shefin, D. Gupta, T. Le, and S. Alqahtani, “Verification-guided falsification for safe rl via explainable abstraction and risk-aware exploration,” 2025.
- [9] J. Zeng, Z. Li, and K. Sreenath, “Enhancing feasibility and safety of nonlinear model predictive control with discrete-time control barrier functions,” in *IEEE Conference on Decision and Control (CDC)*, 2021, pp. 6137–6144.
- [10] Y. Jiang, C. Wang, R. Zhang, J. Wu, and L. Fei-Fei, “Transic: Sim-to-real policy transfer by learning from online correction,” in *Conference on Robot Learning (CoRL)*, 2024. [Online]. Available: <https://arxiv.org/abs/2405.10315>